Design Proposal

Agnes Huang, Riya Affrin, Lirong Que, Runxin Shi

Design Concept

- PlantPal is an AI-powered plant care system that provides proactive, time-based guidance to support novice and busy plant owners.
- Instead of operating as a standalone question-and-answer chatbot, PlantPal incorporates an AI-driven care calendar that helps users anticipate, plan for, and adapt to their plants' ongoing needs.
- PlantPal treats plant care as a longitudinal, situated activity, combining an adaptive care schedule with an initial configuration flow. This shifts users toward sustained, everyday, embedded care.
- PlantPal shows how AI can help with care over time by assisting with planning, adjustments, and forming habits while keeping users informed.

Design Focus

- Our design focuses on a vertical prototype of the interaction with PlantPal's AI-driven care calendar, emphasizing how AI supports planning, adaptation, and habit formation over time.
- We will thoroughly design and implement the interaction flow for initial configuration, time-based care reminders, and schedule modifications. This will demonstrate how AI transitions from reactive question-and-answer to proactive, contextual guidance. The prototype will demonstrate how the AI justifies its suggestions and how users can negotiate, defer, or adjust them.
- We will not aim to implement accurate plant sensing, biological modeling, or real environmental data integration. Instead, these elements will be mocked using predefined plant profiles, simulated conditions, and static inputs. This allows us to focus our design exploration on human–AI interaction, timing, and user agency, rather than technical accuracy.

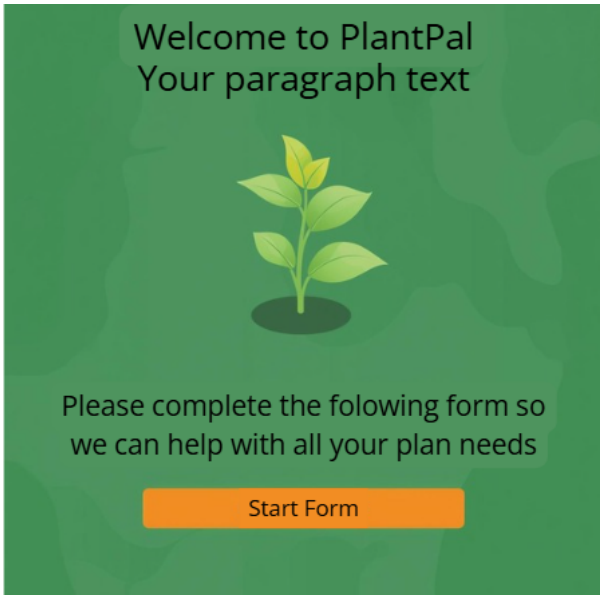
Scenario 1

Task and Story

Identify the user's plant and generate a personalized care plan that includes watering, sunlight, and repotting guidance.

A 21 year old college student downloads the app after buying plants for his apartment but is unsure how to care for them properly. During the initial chatbot-style setup, the system asks simple questions about his living environment, experience level, and care goals. If the user does not know the plant type, he uploads a photo for identification. The app also asks whether he has pets or children and checks for allergies to ensure the plant is safe for his household. Based on this information, the system generates a personalized care schedule with reminders, and the home screen displays a plant image reflecting the plant's current health status..

Walkthrough

	User opens the app and is greeted with a short introduction explaining the gilded plant set up process
[more images]	

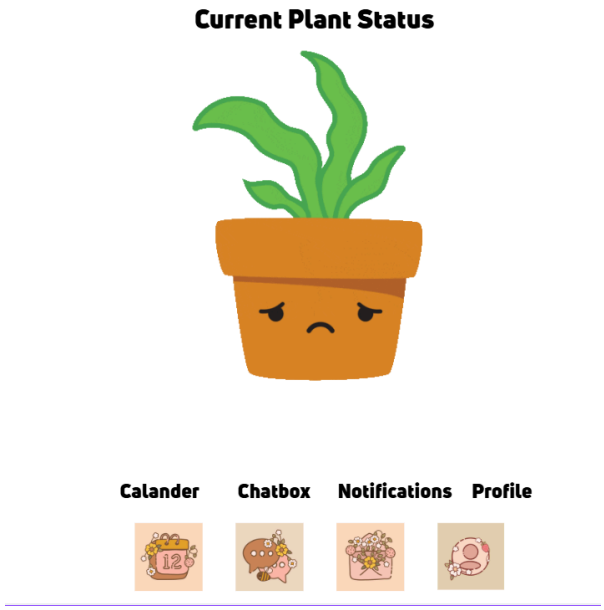
	This section asks the user which city they live in so that it can actively u
[more images]	

What city do you currently live in? Select **Which environment do you live in** • City • Rural • Suburbs	The system asks multiple-choice questions about the user's living environment, and asks about the city it lives in so it has accurate data about weather.
[more images]	

Upload plant image or insert plant name Upload photo or text	The user inputs their plant if known or uploads a photo for automatic plant identification.
[more images]	

Household questions 1. Do you have any pets • Yes • No 2. If yes what kind and how many: 3. Do you have children • Yes • No 4. Do you or anyone else have any allergies worth noting? • Yes • No 6. If yes what are they:	The app asks whether the user has pets or children and checks for allergies to ensure the plant is safe for the household.

[more images]	

	The system generates a customized care schedule and shows a home screen with a plant image reflecting the plant's current health. At the bottom there are links to the users calendar, the AI chatbox for questions, a notifications center and a profile section
[more images]	

Language Model Support

The language model is used to interpret user inputs from multiple-choice selections, optional "Other" fields, uploaded plant photos, and free-form notes. It synthesizes this information to identify the plant, assess household safety, and generate a personalized care schedule. The model also adapts recommendations based on environmental factors and user preferences, and can rearrange tasks if inputs change. Initial exploration with conversational AI tools showed that language models effectively combine structured and unstructured input to produce clear, actionable care guidance.

Scenario 2

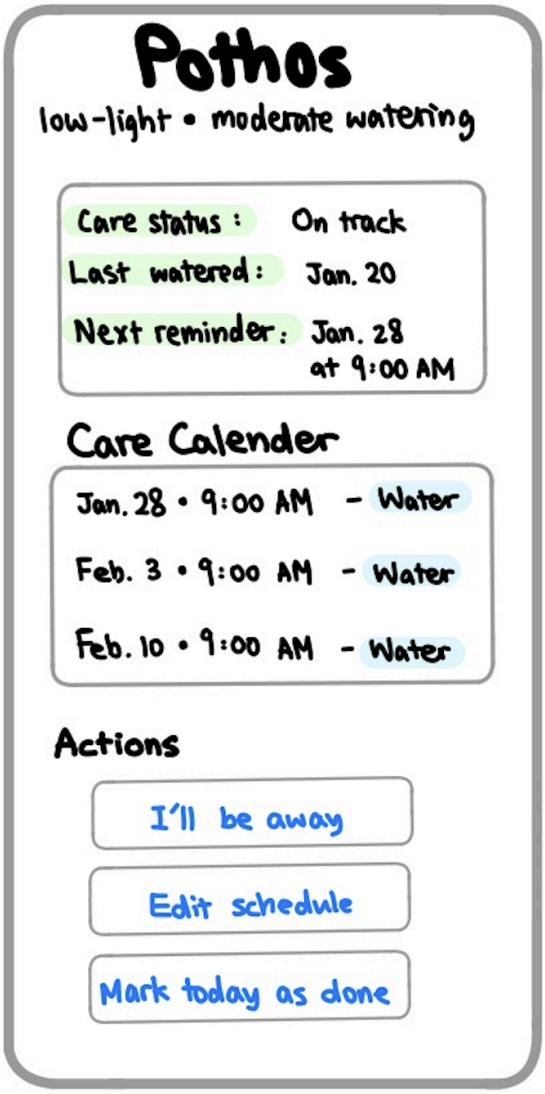
Task and Story

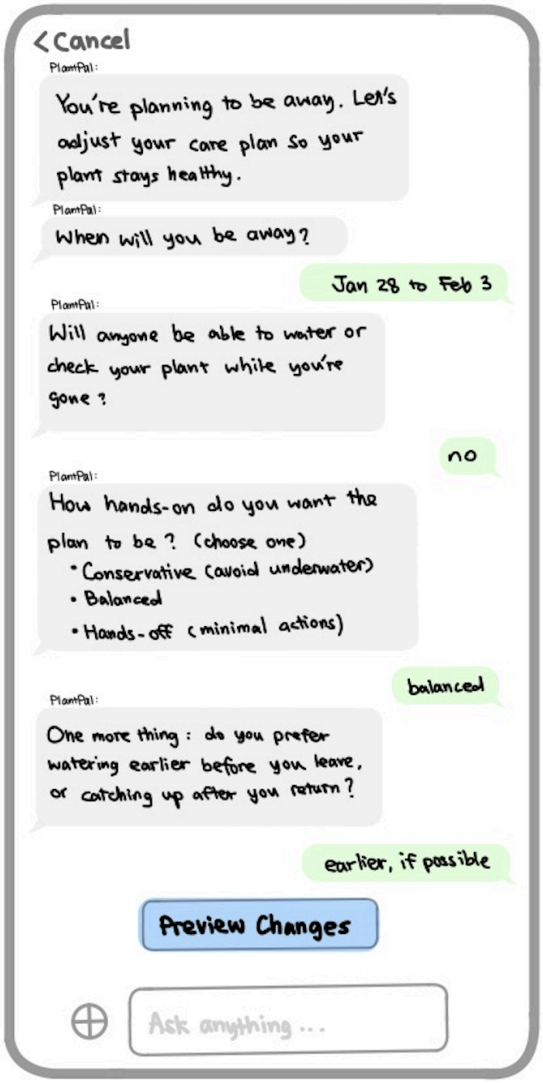
- [This must not exceed 100 words.]
- [See detailed guidance in the assignment.]

Adjust an existing PlantPal care calendar for a 7-day trip by rescheduling reminders and adding a temporary alternative care plan.

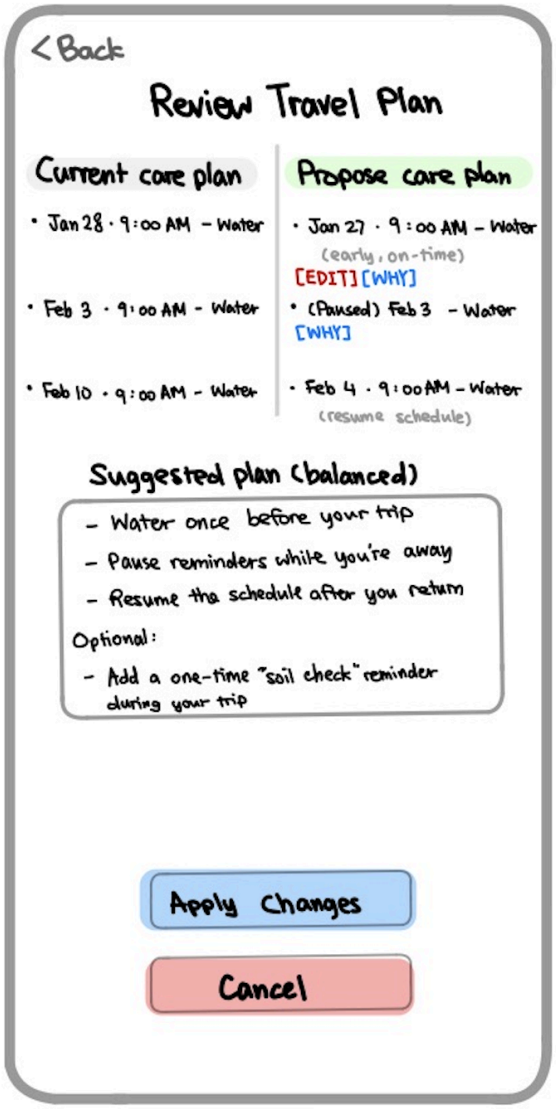
Jamie already follows PlantPal's weekly watering reminders for a pothos. A week before leaving town (Jan 28–Feb 3), Jamie opens "I'll be away," enters the dates, and says nobody can water it. PlantPal previews calendar changes: water one day early, pause reminders during travel, and resume the day after return, plus an optional "ask a friend to check soil" reminder. Jamie edits one suggestion (move early-watering to Saturday morning), then confirms. PlantPal updates the calendar and shows a change summary.

Walkthrough

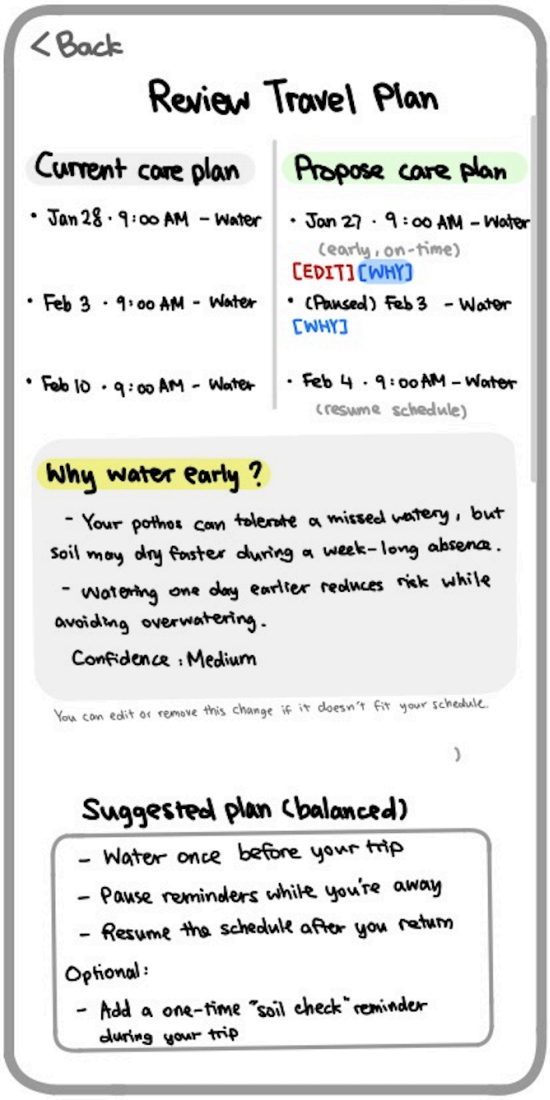
	Jamie opens the plant detail page for their pothos and sees the current care schedule. The system shows that the next watering is scheduled for Jan 28 at 9:00 AM, with future weekly reminders already set. From this page, Jamie can view upcoming care events and access schedule controls. Seeing that an upcoming reminder will occur during their trip, Jamie selects the "I'll be away" option to modify the schedule.



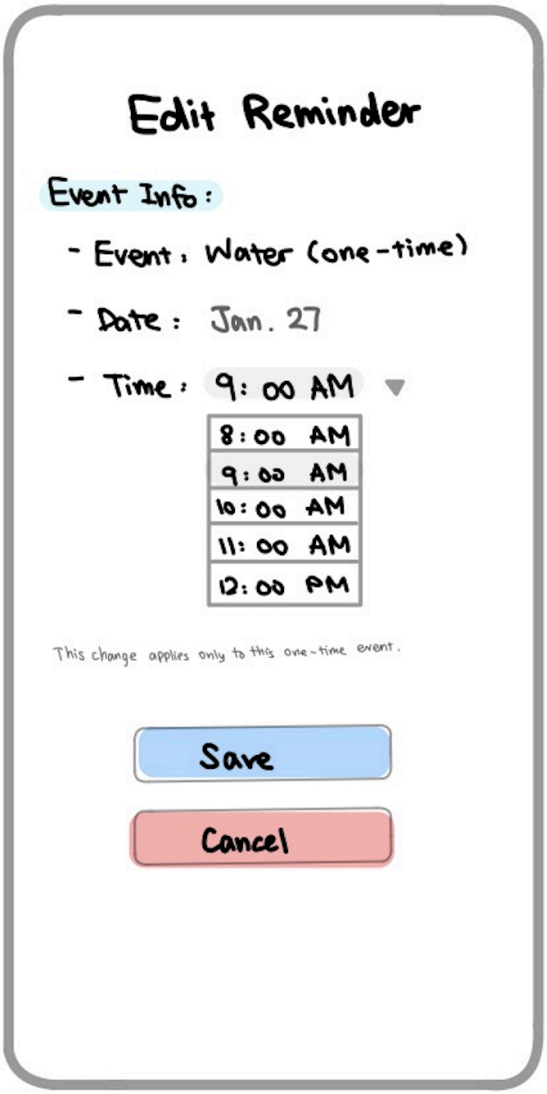
After entering travel mode, Jamie provides the dates of their trip (Jan 28–Feb 3) and answers a small number of follow-up questions. Through short conversational prompts, the system asks whether anyone can help care for the plant and how hands-on the plan should be. These inputs allow the system to understand Jamie's constraints while keeping the interaction lightweight and focused. Then Jamie clicked the "Preview Changes" button.



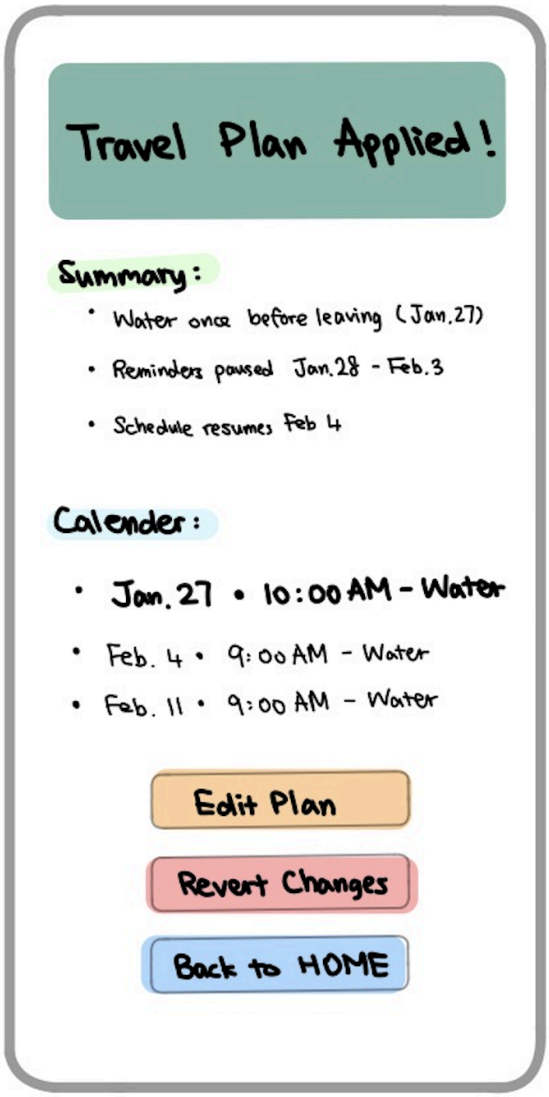
Based on Jamie's input, PlantPal generates a proposed set of concrete schedule changes and presents them for review. The interface shows the current schedule alongside suggested modifications, including moving the Jan 28 watering to Jan 27 as a one-time early action, pausing reminders during the travel period, and resuming the regular schedule on Feb 4. At this stage, the changes are presented as a preview and have not yet been applied.



Jamie taps the “Why” button next to the suggested early watering on Jan 27. A side panel expands to explain the system’s reasoning, including plant-specific considerations and the confidence level of the recommendation. This allows Jamie to understand why the system proposed this change without requiring them to read a long explanation by default.



Deciding to adjust the plan slightly, Jamie selects “Edit” for the one-time early watering event. The system opens a focused editor for that single reminder, allowing Jamie to change the time while keeping the date fixed. This interaction lets Jamie negotiate the system’s suggestion by directly manipulating a concrete calendar event rather than revising the entire plan. Then Jamie clicked the “Save" button.

	After reviewing the suggestions and making a small adjustment, Jamie applies the changes. The system updates the care calendar and confirms that the travel plan is now active. The updated schedule shows the early watering on Jan 27, paused reminders during the trip, and the regular weekly schedule resuming on Feb 4, giving Jamie confidence that their plant care plan has been successfully adapted.

Language Model Support

- [This must not exceed 100 words.]
- [See detailed guidance in the assignment.]

PlantPal uses a language model to interpret a user's travel message (dates and constraints such as "no one can help") and translate it into a structured request to modify an existing care schedule. The model proposes concrete calendar changes (e.g., moving the Jan 28 watering to a one-time early reminder on Jan 27, pausing reminders during Jan 28–Feb 3, and resuming the weekly schedule) and generates short "Why" explanations for each suggestion. The model functions as a proposal generator, and all changes are previewed and can be edited by the user before being applied to the calendar.

Reflection on an HAI Principle

Principle

MI-1: Develop significant value-added automation.

Reflection

PlantPal reflects MI-1 by automating planning decisions that users often find difficult or easy to overlook. Through an initial question-and-answer flow, the system identifies plants that fit a user's lifestyle, experience level, and constraints, reducing guesswork and early plant failure. Once basic contextual information is known, especially future absences or periods of limited availability, PlantPal can proactively adjust and generate a complete care calendar rather than asking users to manually schedule watering, fertilizing, and other tasks. This form of automation adds value by transforming incomplete and informal user input into a concrete, time-based plan that can be followed over time. Working on this prototype also clarified that automation sometimes can be small but also helpful and meaningful. By structuring care around a flexible calendar, the system supports consistent plant care while still allowing users to revise plans as their routines change.

Word Count

To support efficient review, ensure you are within word count limits.

Design Concept: 95 words, limit 100
Design Focus: 126 words, limit 150
Scenario 1 Task and Story: [###] words, limit 100
Scenario 1 Language Model Support: [###] words, limit 100
Scenario 2 Task and Story: 100 words, limit 100
Scenario 2 Language Model Support: 98 words, limit 100
Reflection: 141 words, limit 150